

Contextual Intelligence Construction

Automating the FDE Lifecycle (80:20) for the Modutecture AEC use case, inside StatusNeo's four-loop GCC model

What this is: a use-case scenario showing how the Forward Deployed Engineer's lifecycle automates — 80% by an agentic AI swarm, 20% by the human Pod Commander — applied to **Modutecture's “Continuous Contextual Construction”** platform, and mapped onto StatusNeo's Clarity → Build → Velocity → Improvement loop. This is a scoping artifact for the charter, not a delivery commitment.

STATUS & SOURCING NOTE

This scenario integrates June-2026 claims from the Modutecture blueprint and StatusNeo GCC material: specific tools (Perspective AI, OpenSpace, the StatusNeo™ accelerators) and metrics (14-day TTFV, 85% margin, 39-test gate). **Treat them as the target operating model, pending verification — not validated results.** Durable: the loop structure + 80:20. Perishable: the named tools and numbers.

The problem the FDE is dropped into

Modutecture builds a “Continuous Contextual Construction” platform for the AEC sector — an **Agentic Twin** that reasons across building codes, BIM models, and live site/sensor data, targeting high-compliance verticals (e.g., the Kaiser Permanente healthcare context). The hard part is not rendering a model; it is keeping the twin *correct and code-compliant* as reality changes — where clinical safety and building-code compliance are non-negotiable.

The FDE's job: embed in that environment and stand up the agentic pipeline that takes a stakeholder's intent → a code-compliant, twin-verified construction artifact — automating the mundane 80% and reserving the 20% (architecture, compliance sign-off, client trust) for the human.

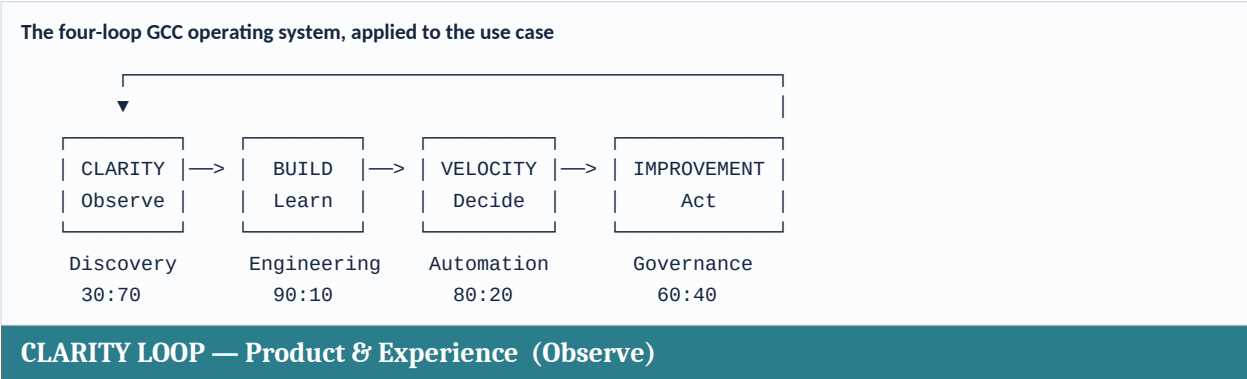
The 80:20 framing for this use case

In 2026 the FDE no longer manually wires APIs or writes glue code. They orchestrate PODs that autonomously handle the 80% of mundane execution, leaving the 20% of high-stakes strategic reasoning and safety sign-off to the human — critical in a domain where an unsafe artifact has physical consequences.

AI 80%		HITL 20%	
The AI swarm owns (80%)		The human Pod Commander owns (20%)	
BIM federation, GraphRAG grounding over codes, draft generation, test authoring, site-to-twin diffing.		The architecture of the twin, the compliance interpretation, the clinical-safety sign-off, client trust.	
Running the 14-day velocity loop to first verified artifact.		Deciding the artifact is safe to act on — the irreversible call.	

PART II • EXECUTION ACROSS THE FOUR LOOPS

StatusNeo's GCC operates as a four-loop system — Observe → Learn → Decide → Act, expressed as Clarity → Build → Velocity → Improvement. Below, the Modutecture use case is executed loop by loop, with the AI:Human split and the proprietary accelerators (TestCraft™, QueryButler™, NeoLens™, Backstage) shown per loop.



Capabilities in this loop: Experience & interface design, product management & strategy, AI product design.

AI swarm (30%): Perspective AI runs stakeholder discovery; ingests building codes, BIM topology, and site context into a GraphRAG knowledge layer; drafts the candidate scope and the spec.

Human FDE (70%): The FDE frames the real problem with the client, validates the scope against clinical/operational reality, and authors the intent the swarm will build to.

Gate: Human approves the spec and scope before any build begins.

BUILD LOOP — Engineering (Learn)



Capabilities in this loop: Product engineering, data & AI engineering, platform automation engineering.

AI swarm (90%): The agent swarm performs BIM federation, generates the twin logic and the construction artifact via Spec-Driven Development, and self-corrects against the GraphRAG-grounded codes.

Human FDE (10%): The FDE makes the architecture calls and spot-reviews novel structural/clinical logic; otherwise stays out of the swarm's way.

Gate: Architecture/contracts frozen as ADRs; output flows to the velocity loop.

VELOCITY LOOP — Automation (Decide)

AI 80%

HITL 20%

Capabilities in this loop: DevSecOps, SRE & CloudOps; test engineering & automation.

AI swarm (80%): TestCraft™ + the Adversarial SDET run the correctness gate (the 39-test compliance suite per source); OpenSpace performs site-to-twin visual verification; canary/rolling deploys run within guardrails.

Human FDE (20%): The FDE holds the ship gate — the decision that a code-compliant, twin-verified artifact is safe to release into a clinical environment.

Gate: THE CORRECTNESS GATE — 100% test pass + compliance suite + human safety sign-off.

IMPROVEMENT LOOP — Governance & Scale (Act)

AI 60%

HITL 40%

Capabilities in this loop: AI & engineering governance; compliance, security & risk; observability & maturity index.

AI swarm (60%): NeoLens™ (AIOps) monitors the live twin, detects drift between site and model, and re-enters the loop; QueryButler™ keeps the knowledge/data layer current.

Human FDE (40%): The FDE owns the governance posture, tunes the guardrails and the maturity index, and owns the post-incident call — then decides where to scale next.

Gate: Auto-heals within policy; escalates to the human on drift or repeated failure.

The specific 2026 tools (per source) mapped to the lifecycle and the 80:20 split. *Named tools and metrics are source-attributed and pending verification.*

Lifecycle stage	Tool (per source)	Role in the use case	Split
Discovery	Perspective AI	Stakeholder discovery; intent capture into GraphRAG.	30:70
Knowledge/data	QueryButler™	Grounds the swarm in building codes + BIM topology.	85:15
Engineering	Spec-Driven Dev + swarm	BIM federation + twin-logic generation.	90:10
Quality gate	TestCraft™ + Adversarial SDET	39-test correctness/compliance suite.	80:20
Visual verify	OpenSpace	Site-to-twin visual verification (reality vs. model).	85:15
Dev portal	Backstage	Golden paths, policy-as-code, self-service governance.	80:20
Observability	NeoLens™ (AIOps)	Live twin monitoring + drift detection + self-heal.	80:20

The deterministic gate — why this use case needs it

In a clinical/AEC context, the agentic speed (the 14-day loop per source) must never compromise integrity. The **deterministic correctness gate** is the non-negotiable: an artifact does not pass on an agent's say-so — it passes a compile + a compliance test suite (39 tests per source) + a human safety sign-off. **This is the 80:20 made safe: the AI proposes the artifact; the human disposes on whether it is safe to build.**

THE GOVERNING PATTERN FOR THIS USE CASE

Speed without a gate is a liability in a hospital. The agentic swarm gives the FDE the velocity of a full team; the deterministic correctness gate plus the human safety sign-off give the accountability a clinical build requires. AI proposes the contextual artifact; the human disposes on safety. That gate is the 20% — and in this domain, it is the whole point.

The Modutecture engagement is the **tip of the spear**: a single elite Pod Commander backed by an autonomous AI swarm proves the contextual-construction motion and secures architectural trust — before it scales into a full StatusNeo GCC. This is the land-and-expand logic of the whole FDE practice, applied to one AEC account.

Phase	What happens	Topology
Land (FDE)	The Pod Commander embeds, runs the four loops on the Modutecture twin, hits first verified artifact, earns trust.	Embedded — 70:30
Prove	The correctness gate + site-to-twin verification demonstrate safety to the client's compliance teams.	Embedded
Expand (GCC)	Once trusted, the offshore GCC pods scale delivery behind the FDE using the same loop OS and accelerators.	Remote — 90:10

The 90-day arc (per StatusNeo GCC model): Day 0–15 blueprint + landing zone; Day 16–45 pods live + portal active; Day 46–90 operate, measure, optimize — the GCC goes from slideware to running software.

Open scoping questions (to resolve in the charter)

1. Which Modutecture loop is the pilot's first proof point — the Build loop (twin generation) or the Velocity loop (the correctness gate)?
2. What is the real composition of the “39-test correctness gate” for the Kaiser/clinical context, and who owns its definition — Modutecture, the client's compliance team, or the FDE?
3. Where does authoritative twin geometry live — in the Unity/BIM layer or the twin backend? (The decisive test: can a non-Unity client reconstruct a Moducule from backend data alone?)
4. Local compute (RTX 4090 nodes per source) vs. cloud G7 — which substrate for the Adversarial SDET's fuzzing load, given air-gapped/data-residency constraints?
5. Which named tools are real and licensed today vs. aspirational — Perspective AI, OpenSpace, and the StatusNeo™ accelerators all need a build/buy/verify status before they enter the scope.

THE ONE-LINE USE CASE

Embed a Pod Commander + AI swarm in Modutecture's contextual-construction platform; automate 80% of the twin pipeline and reserve the 20% (architecture, compliance, clinical safety sign-off) for the human; prove it on one clinical account through a deterministic correctness gate; then scale into a StatusNeo GCC.